

What report design does

Report View: Turn prepared data and DAX logic into visuals users can understand

↓
Choose visuals based on the business question

Choose the Right Visual

Match visual to purpose

- ✓ Trend: line chart
- ✓ Comparison: bar or column
- ✓ KPI: card or KPI visual
- ✓ Distribution: scatter or histogram

Format Clearly

Improve readability

- ✓ Clear titles on every visual
- ✓ Rename axes and labels
- ✓ Format numbers properly
- ✓ Use tooltips for extra context

Add Analytical Context

Help users interpret results

- ✓ Average and median lines
- ✓ Trend and forecast lines
- ✓ Reference and percentile lines
- ~ Use only when adding insight

Key questions before finalising visuals

What business question does this answer?

Is the right visual type chosen?

Is it a report or a dashboard?

Are titles and labels clear?

Are numbers formatted correctly?

Does the design support the insight?

"A visual should be chosen because it answers a question, not because it looks impressive."

Next up: Week 8 - Interactive Storytelling

Introduction: What is Report Design?

Once data has been connected, cleaned, modelled, and enhanced with DAX, the next step is to present it visually. In Power BI, report design is the process of choosing the right visuals, arranging them clearly, formatting them properly, and making sure the report answers the business question. Visuals are the core elements of a Power BI report. They allow users to explore data, compare results, track KPIs, identify trends, and understand patterns after the data has been prepared and modelled.

In simple terms: **Report design is where data becomes understandable.**

A report should not only show numbers. It should help users interpret those numbers and decide what to do next.

What Are Visuals and Why Are They Important?

Visuals are charts, tables, cards, maps, slicers, and other report elements used to represent data inside Power BI.

Power BI User Interaction Sequence

Understand Performance at a Glance
Users get a quick overview of business performance

Track KPIs and Targets
Users monitor key performance indicators and targets

Identify Outliers and Patterns
Users detect unusual data points and recurring patterns

Filter and Segment the Report
Users refine data by filtering and segmenting

Compare Categories
Users can compare different categories of data

Analyse Trends Over Time
Users examine trends and patterns in data over time

Explore Relationships Between Variables
Users investigate connections between different variables

Support Storytelling and Decision-Making
Users use data to tell stories and make informed decisions

Visuals can also support exploratory data analysis. Analysts can use them to inspect trends, understand distributions, identify relationships, and even support feature selection before machine learning or deeper modelling work.

Power BI includes many built-in visuals, such as cards, tables, matrices, bar charts, line charts, maps, slicers, KPI visuals, and more advanced visuals such as decomposition tree, Q&A, and smart narrative. Analysts can also access custom visuals from AppSource when standard visuals do not meet the reporting need.

The key idea is: **A visual should be chosen because it answers a question, not because it looks impressive.**

Choosing Visuals Based on Purpose

Different visuals serve different analytical purposes.

 Choosing Visuals Based on Purpose Different visuals serve different analytical purposes.		
 Purpose	 When to Use	 Suitable Power BI Visuals
 Single Number	When users need one key metric at a glance	 Card  KPI card  multi-row card
 Goal Tracking	When comparing actual performance against a target	 KPI visual  gauge  card with conditional formatting
 Comparison	When comparing values across categories	 Bar chart  column chart  stacked bar/column chart
 Trend Over Time	When showing movement across dates or periods	 Line chart  area chart  combo chart
 Part-to-Whole	When showing contribution to a total	 Donut chart  pie chart  stacked bar  100% stacked chart
 Ranking	When showing top or bottom performers	 Bar chart  table  matrix with sorting
 Detailed Table View	When users need structured row-and-column detail	 Table  matrix
 Spatial Analysis	When location or geography matters	 Map  filled map  Azure Maps
 Distribution	When showing spread or frequency of values	 Histogram  column chart with bins  box-style custom visuals
 Correlation	When comparing two numeric variables	 Scatter chart
 Flow or Breakdown	When showing contribution, hierarchy, or root cause paths	 Waterfall chart  decomposition tree
 Filtering	When users need to narrow the report view	 Slicer  filter pane
 Narrative Summary	When users need written insight support	 Smart narrative  text boxes

A simple rule: **Start with the business question, then choose the visual.**

Visual Beautification and Formatting

Good formatting improves readability and user confidence. Power BI visuals can be improved through:

Axis Naming: Rename axes so users understand what the visual is showing. Avoid unclear technical field names. Example: Instead of SalesAmount, use **Total Sales**.

Titles: Every visual should have a clear and meaningful title. Example: Instead of **Sales Chart**, use **Monthly Sales Trend by Region**.

Data Labels: Use data labels when users need exact values, but avoid cluttering the visual.

Legends: Use legends carefully when multiple categories are being compared.

Background and Effects: Use background, borders, and shadows sparingly. The design should support the insight, not distract from it.

Gridlines: Use gridlines when they improve readability. Remove them when they make the visual look crowded.

Tooltips: Tooltips provide extra context without overcrowding the report page. Example: A bar chart may show sales by product, while the tooltip shows profit margin, quantity sold, and target variance.

Data Formatting: Format numbers properly using currency, percentages, decimals, and units. Example: Use **\$1.2M** instead of **1200000**.

The goal is to make visuals easier to read, not just more colourful.

Adding Analytical Context to Visuals

Power BI also provides analytical options that help users interpret the data more clearly. [Note: All the listed features may not be available for all kinds of visual. The analytical context depends on the type of chart chosen]

These can include:

- Average line
- Minimum line
- Maximum line
- Median line
- Percentile line
- Trend line
- Forecasting
- Error bars
- Reference lines

These features help users compare values against benchmarks, spot patterns, and understand whether a result is unusual.

Example: A sales trend chart with an average line can quickly show whether recent performance is above or below the usual level. Analytical features should be used when they add context to the business question.

Report vs Dashboard

In Power BI, reports and dashboards are related, but they are not the same. A **report** is an interactive analysis experience. It can have one or more pages and usually includes multiple visuals, slicers, filters, and drill paths. Reports can be created in Power BI Desktop and Power BI Service. A **dashboard** is a single-page summary view created in Power BI Service. It is built by pinning visuals from reports, datasets, or other content.

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Report vs Dashboard

Understand the differences between reports and dashboards in Power BI.

 Area	 Report	 Dashboard
 Created in	Power BI Desktop or Power BI Service	Power BI Service
 Pages	Can have multiple pages	Single page
 Interactivity	Highly interactive	More limited
 Purpose	Detailed analysis	High-level monitoring
 Built from	Dataset / semantic model	Pinned visuals or tiles
 Best used for	Exploration and reporting	Executive overview or monitoring

A report helps users explore. A dashboard helps users monitor.

Practical Way to Decide



A Simple Decision Guide for Power BI Visual Selection

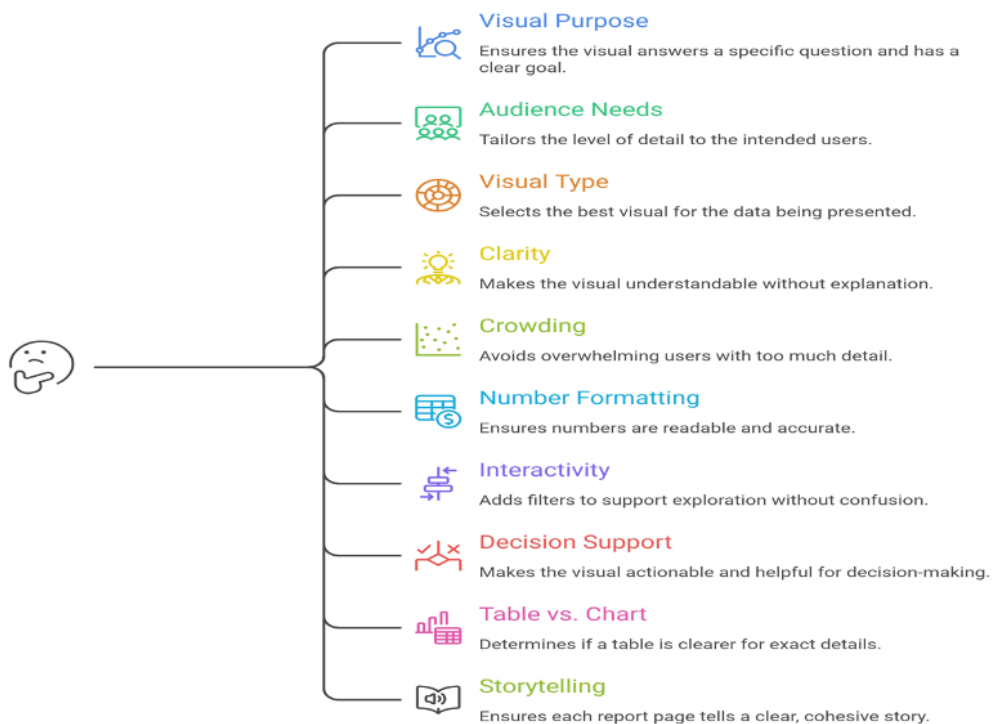
Choose the right visual based on the business question you are solving.

1 CARD		Use a Card when: The user needs one important number at a glance.
2 KPI		Use a KPI visual when: The user needs to compare performance against a target.
3 BAR / COLUMN		Use a Bar or Column Chart when: The user needs to compare categories.
4 LINE CHART		Use a Line Chart when: The user needs to understand change over time.
5 MATRIX		Use a Matrix when: The user needs structured detail across rows and columns.
6 MAP		Use a Map when: Location adds meaning to the analysis.
7 SLICER		Use a Slicer when: The user needs to filter the report interactively.
8 TOOLTIPS		Use Tooltips when: The user needs extra detail without overcrowding the page.
9 ANALYTICAL LINES		Use Analytical Lines when: The user needs benchmarks, averages, minimums, maximums, or trends.
10 DASHBOARD		Use a Dashboard when: The goal is high-level monitoring in Power BI Service.
11 REPORT		Use a Report when: The goal is interactive analysis and deeper exploration.



The right visual helps turn data into clear and actionable insights.

What Analysts Should Ask Before Finalising Visuals



Conclusion

Report design is where Power BI turns prepared data, strong models, and DAX logic into insights that users can understand and act on. Choosing the right visual is not about making the report look attractive; it is about matching the visual to the business question, the audience, and the decision being supported.

A good report helps users see what matters quickly, explore details when needed, and trust the story behind the numbers.

Stay tuned for the next article in **The Insight Stack: Power BI, Layer by Layer**, where we move into **Week 8: Interactive Storytelling** and explore how analysts use drill-through, tooltips, bookmarks, navigation, and user interactions to guide decision-making inside Power BI.

Further Readings

- Visualizations Overview: [Overview of visualizations in Power BI - Power BI | Microsoft Learn](#)
- Card Visuals: [Create a card visual in Power BI - Power BI | Microsoft Learn](#)
- KPIs: [Key Performance Indicator \(KPI\) visuals - Power BI | Microsoft Learn](#)
- Matrix: [Create a Matrix Visual in Power BI - Power BI | Microsoft Learn](#)
- Column Charts: [Create and use column charts in Power BI - Power BI | Microsoft Learn](#)
- Line Charts: [Line Charts in Power BI - Power BI | Microsoft Learn](#)
- Maps: [Overview of Map Visualizations in Power BI - Power BI | Microsoft Learn](#)
- Slicers: [Overview of slicers in Power BI - Power BI | Microsoft Learn](#)